

IN COMMAND WINDOW:

```
addpath /Users/megsheppard/Documents/MATLAB/spm12
```

```
cd /Users/megsheppard/Desktop/vbm_spm
```

```
spm pet
```

```
//
```

First: Check that all files are in .nii (nifti format)

- This combines the actual image data (originally in .img) and the information necessary for SPM to understand it (originally in .hdr)
- If they're not you need to convert them from .dicom to .nii

DISPLAY → select an image: it will show three slices through the image. If it is in the correct format, then:

- Top left is coronal (superior of the head displayed at top, left shown on left – as if the subject is viewed behind)
- Bottom left is axial (anterior of the head displayed at top, left shown on left – as if the subject is viewed from above)
- Top right image is sagittal (anterior of the head at the left – as if the subject is viewed from the left)

PREPROCESSING FOR VBM

This is different when you're utilising a large number of files (as you do in a VBM)

1. SEGMENTATION

AIM: to identify different types of tissues within the image (grey matter, white matter, CSF, bone etc).

A. SPM Gui → Batch

This opens the Batch Editor Window

B. Choose SPM → Spatial → Segment

This will open a new window – you will only need one 'channel'

C. Volumes → Select all your data files

- a. Bias regularisation: leave as default
- b. Bias FWHM leave as default
- c. Save bias corrected: **save nothing**

D. Tissues → there are five different subsections titled tissue.

For all 5 always leave 'Tissue probability map', 'number of gaussians' and 'warped tissue' as default.

Tissues → first tissue corresponds to grey matter

Alteration → Native Tissue: change to 'Save Native + DARTEL imported'

Tissues → second tissue corresponds to white matter

Alteration → Native Tissue: change to 'Save Native + DARTEL imported'

Tissues → third tissue corresponds to CSF

Alteration → Native Tissue: change to 'Native Space'

Tissues → fourth tissue corresponds to skull

Alteration → Native Tissue: change to 'none'

Tissues → fifth tissue corresponds to air

Alteration → Native Tissue: change to 'none'

Everything else is leave as default and run.

OUTPUT:

Files containing c1 ← identified as grey matter

Files containing c2 ← white matter

Files containing c3 ← CSF

Files beginning with 'r' e.g., **rc1** ← DARTEL imported versions of tissue class images (to be aligned next)

SANITY CHECK:

Check Reg → check resulting images

- For one of the subjects select the original, the c1, c2 and c3. This should give an idea about which voxels the algorithm identifies as different tissue types.

2. RUN DARTEL (create templates)

- This increases the accuracy of inter-subject alignment
- Models shape of each brain using millions of parameters (3 per voxel)
- This works by aligning the grey matter among the images – simultaneously aligning white matter
- Generates crisp average template data – uses the imported 'rc1' and 'rc2' images – **generates "u_rc1"** files (and a series of template images)

A. SPM Gui → Batch

This opens the Batch Editor Window

B. Choose SPM → Tools → DARTEL Tools → create DARTEL (+templates)

→ images: two channels of image need to be created

- IMAGES: select the imported grey matter images → **rc1*.nii**
- IMAGES: select the imported white matter images → **rc2*.nii**
 - Need to be in the same specified order so they correspond
- SETTINGS: leave as default

this takes a long time to run – be patient

3. NORMALISE TO MNI SPACE

Uses the u_rc1 files to generate smoothed, spatially normalised and Jacobian scaled grey matter images in MNI space.

A. SPM Gui → Batch

This opens the Batch Editor Window

B. Choose SPM → Tools → DARTEL Tools → Normalise to MNI Space

→ DARTEL TEMPLATE: select the final template image (Template_6.nii) – registered to MNI space (affine transform) – allowing the transformations to be combined so that all the individual spatially normalised scans can be brought into MNI space

→ SELECT ACCORDING TO: choose many subjects

○ MANY SUBJECTS

- FLOW FIELDS: select all the flow fields created by the previous step (u_*.nii)
- IMAGES: need one channel of images if only analysing grey matter
 - IMAGES: select all the grey matter images (c1*.nii) in the same order as the flow fields
- VOXEL SIZES: specify voxel sizes for spatially normalised images – Leave as is to have 1.5mm voxels.
- BOUNDING BOX: leave as default
- PRESERVE: for VBM set to preserve amount (“modulation) – compares tissue volumes
- GAUSSIAN FWHM: 10mm [choose between 4 and 12, I like 6 personally]

save in JOBS folder and press GO

OUTPUT: .mat files (or .m)

STATISTICAL ANALYSIS

Time to fit the GLM model through the pre-processed data

Before you do any of this **set up the excel sheet required for covariates (i.e., see the Global Calculation Sheet)**

SPM Gui:

Basic Models → opens Batch editor window with a '*Factorial design specification*' (this is the default, so it does not need changing)

1. Directory
 - Set working directory
2. Design
 - Change from 'one-sample t-test' to 'multiple regression'
 - Scans:
 - Select "smwc1" files
 - Intercept:
 - Specify Menu Item → Include Intercept
 - Covariates:
 - Select 'New Covariate' x3 (controlling for sex and age and your variable of interest)
 - Enter "info(:,v)" as Vector "ELS" as name and No centering for Centering
 - Enter "info(:,x)" as the Vector, "Sex" as the Name and No centering for Centering. Values of 0 indicate a female subject, and 1 indicates male.
 - Enter "info(:,y)" as the Vector, "Age" as the Name and No centering for Centering

This is the information from the excel sheet you've created and imported where x and y correspond with the rows in the csv file

3. Masking: determines which voxels will be included in the analysis
 - Multiple comparisons corrections are more sensitive if fewer voxels are included

USE Absolute Masking: threshold of 0.2

4. Covariates: we do not add our covariates here, they're specified in the design of the model
 - Keep this empty
5. Global Calculation → you have already set all of this up...
 - Change to 'User'
 - Then input $\text{info}(:,z)$ ← where z represents the row in the csv for your TCV for each participant
 - Alternatively you can sum the TCV here by stating $\text{info}(:,4) + \text{info}(:,5) + \text{info}(:,6)$
6. Global Normalisation → indicates how the globals should be used
 - a. Overall grand mean scaling: no
 - b. Normalisation:
 - a. None: indicates no global correction should be done
 - b. Proportional scaling – indicates data should be divided by the globals
 - i. Scales the values in the pre-processed data so they're proportional to the fraction of brain volume accounted for
 - c. Ancova correction treats globals as covariates in the GLM
 - i. Results will show differences that cannot be explained by the globals

Proportional scaling is recommended here!

press GO

OUTPUT: SPM.mat file

REVIEW: check to see what has been specified, see any mistakes, and edit!

ESTIMATE: this runs the analysis

- Select the SPM.mat file that has been created
- WAIT this will take a while.

Step one: fit the data at each voxel to some linear combination of the columns in the design matrix

- This generates a series of **beta images**
- First indicates the contribution of the first column, second of second etc.
- A **ResMS image is generated**, providing the necessary standard deviations needed to compute the t statistic

Step two: compute residual images (smoothness of data is contributed)

- Temporary and deleted ASAP

RESULTS:

- **Now the GLM has been fitted, use the beta and ResMS images and the SPM.mat file to generate t-statistics.**
- Contrast vectors are specified indicating a linear combination of beta images to test
 - These are saved as con images – identifying where regions significantly differ from zero
 - T-statistics are saved in “**spmT**” images
 - Gaussians Random Field Theory was used to correct multiple comparisons problem

→ RESULTS:

- c. Select “SPM.mat”
- d. Brings up Contrast Manager
 - a. T-contrasts
 - i. Define new contrast: Age
 - ii. 0, -1, 0
 - iii. OK
 - iv. Done

POP UP:

- e. mask with other contrast(s): click no
- f. title for comparison: leave this as it is. You can hit return on the keyboard or click the edge of the text box with your mouse to do this.
- g. p value adjustment to control: Click FWE (family-wise error correction).
- h. threshold {T or p value}: Enter 0.05 for regions that are significant after corrections for multiple comparisons ($p < 0.05$).

- i. & extent threshold {voxels}: let's see if there are any blobs at all, so go for 0.